CBSE CLASS 12 CHEMISTRY

Most Important Questions Analysis • By Pragya

Analyzing Papers: Set 56/5/1 (Series RP5PS/5), Set 56/7/1 (Series WYXZ7), and Set 56/5/1 (Series HFG1E/5)

1. Physical Chemistry: Numerical & Reasoning

Chemical Kinetics (First Order Reaction)

- **Question:** Show that for a first-order reaction, the time required for **99% completion is twice** the time required for the completion of **90%** of the reaction.

  Sources: 56-5-1_Chemistry.pdf, 56-7-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

- **Question:** Calculate the **Activation Energy (E_a)** given rate constants at two different temperatures, or if the rate quadruples with a specific temperature rise (e.g., 300K to 320K).

  Sources: 56-5-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

Electrochemistry

- **Question:** Calculate the **emf of the cell (E_{cell})** and **ΔG°** for a cell reaction (e.g., $Zn|Zn^{2+} || H^+|H_2$ or $Ni|Ni^{2+} || Ag^+|Ag$) using the **Nernst Equation**.

  Sources: 56-5-1_Chemistry.pdf, 56-7-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

- **Question:** State **Kohlrausch's Law** and calculate molar conductivity (Λ_m) or degree of dissociation (α) given conductivity (κ) and concentration.

  Sources: 56-5-1_Chemistry.pdf, 56-7-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

Solutions

- **Question:** Calculate the mass of solute required to lower the freezing point by a certain amount (Depression in Freezing Point), often involving the **Van't Hoff factor (i)** for electrolytes like $MgCl_2$ or K_2SO_4, or association of acetic acid in benzene.

  Sources: 56-5-1_Chemistry.pdf, 56-7-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

2. Inorganic Chemistry: Theory & Structures

Coordination Compounds

- **Question:** Explain the **color of complexes** (e.g., $[Ti(H_2O)_6]^{3+}$) using **Crystal Field Theory (CFT)**. Why does the color change or disappear on heating?

  Sources: 56-7-1_Chemistry.pdf, 56-5-1_Chemistry.pdf

- **Question:** Using **Valence Bond Theory**, compare the geometry and magnetic behavior of complexes like $[Ni(CO)_4]$ (tetrahedral, diamagnetic) vs $[Ni(CN)_4]^{2-}$ (square planar, diamagnetic).

  Sources: 56_5_1_Chemistry.pdf, 56-5-1_Chemistry.pdf

- **Question:** Identify **Linkage Isomerism** in complexes containing ambidentate ligands (e.g., $[Co(NH_3)_5(NO_2)]Cl_2$).

  Sources: 56-7-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

d- and f-Block Elements

- **Question:** Explain the cause and consequences of **Lanthanoid Contraction** (e.g., why Zr and Hf have similar atomic radii).

  Sources: 56-5-1_Chemistry.pdf, 56-7-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

- **Question:** Account for the E° values of the 3d series: Why is $E°(Mn^{2+}/Mn)$ highly negative or why is Mn^{3+} a strong oxidizing agent?

Sources: 56-5-1_Chemistry.pdf, 56-7-1_Chemistry.pdf

3. Organic Chemistry: Mechanisms & Reactions

Reactivity Orders

- **Question:** Arrange haloalkanes in increasing order of reactivity towards **S_N1 or S_N2** reactions. (Focus on: Tertiary halides for S_N1, Primary for S_N2, and the effect of solvent polarity).

  Sources: 56-5-1_Chemistry.pdf, 56-7-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

- **Question:** Arrange in increasing order of **Acidic character** (Phenols vs Alcohols, effect of -NO_2 groups) or **Basic character** (Amines in aqueous phase).

  Sources: 56-5-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

Name Reactions (Direct or Conversions)

- **Cannizzaro Reaction:** Write the reaction for an aldehyde with no α-hydrogen (e.g., HCHO or Benzaldehyde) with conc. alkali.

  Sources: 56-5-1_Chemistry.pdf, 56_5_1_Chemistry.pdf, 56-7-1_Chemistry.pdf

- **Sandmeyer's Reaction:** Conversion of diazonium salts to haloarenes/nitriles.

  Sources: 56-5-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

Distinction Tests

- **Question:** Give a chemical test to distinguish between:
  - Aldehyde vs Ketone (Tollen's or Fehling's test).
  - Methyl Ketone vs others: **Iodoform Test** (Positive for compounds with CH_3-CO- or CH_3-CH(OH)- groups).
  - Amines: Hinsberg Test (1° vs 2° vs 3°).

Sources: 56-5-1_Chemistry.pdf, 56-7-1_Chemistry.pdf, 56_5_1_Chemistry.pdf

4. Biomolecules

- **Question:** Write the reactions of **D-Glucose** with:
    1. **HCN** (shows carbonyl group).
    2. **Bromine water** (shows aldehyde group).
    3. **HI** (shows straight carbon chain).

Sources: 56-7-1_Chemistry.pdf, 56_5_1_Chemistry.pdf